

Flanker Task EEG — Functional Data Analysis

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Introduction

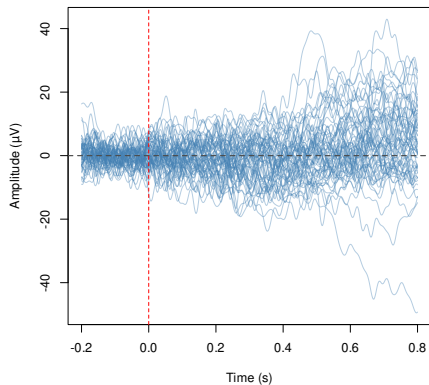
Introduction

- **Dataset:** EEG recordings from 62 young adults performing a Flanker cognitive task (OpenNeuro ds006018)
- **Channel:** FC1 (frontal-central), stimulus S2, time window $[-0.2\text{ s}, 0.8\text{ s}]$
- **Research questions:**
 - Does ADHD status affect the functional EEG response curve?
 - Does gender affect the functional EEG response curve?
 - Does socioeconomic status (SES) affect the functional EEG response curve?
- **Methods:** B-spline smoothing, functional EDA, hypothesis testing, function-on-scalar regression

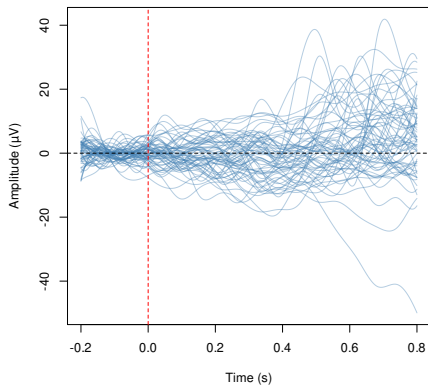
Project Overview

Smoothing

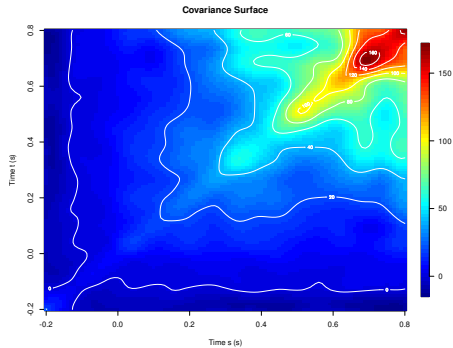
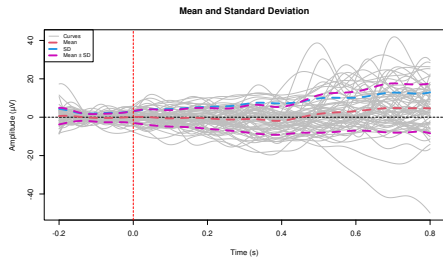
Before Smoothing (raw)



After Smoothing ($\lambda = 10^{-5}$)

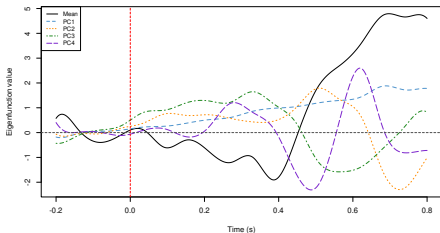


EDA

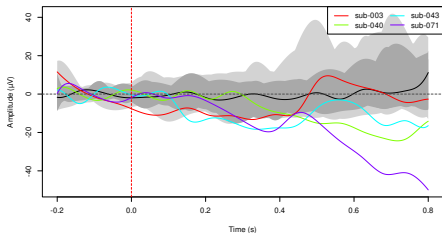


EDA

PCA Harmonics and Mean Function



PC1: 61.1% PC2: 13.6% PC3: 11.9% PC4: 3.3%

HDR Functional Boxplot, $\alpha = (0.05, 0.50)$ 

Magnitude outliers identified: sub-003, sub-040, sub-043, sub-071 — retained in subsequent analyses.

Hypothesis Testing — Summary

$$H_0 : \mu_A(t) = \mu_B(t), \quad \forall t \in [-0.2, 0.8]$$

$$H_1 : \mu_A(t) \neq \mu_B(t), \quad \text{for some } t$$

Comparison	Test	p-value
ADHD vs non-ADHD	L2-norm (bootstrap)	0.7260
	F-type (bootstrap)	0.7140
	Permutation	0.5250
Female vs Male	L2-norm (bootstrap)	0.2600
	F-type (bootstrap)	0.2680
	Permutation	0.1050
SES (Low/Mid/High)	Functional ANOVA	0.4020

Conclusion: All three hypotheses fail to reject H_0 at $\alpha = 0.05$ — no significant differences found between groups.

Regression Model Results

Regression Model

To further investigate the relationship between EEG response curves and subject characteristics, a function-on-scalar regression model was fitted. The regression was performed on the 51 subjects with known ADHD classification, excluding the 11 subjects with missing ADHD status. The model is defined as:

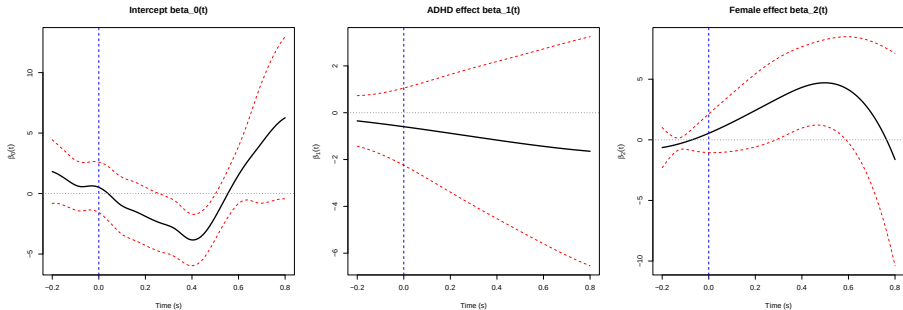
$$Y_i(t) = \beta_0(t) + \beta_1(t) \cdot \text{ADHD}_i + \beta_2(t) \cdot \text{Female}_i + \varepsilon_i(t), \quad t \in [-0.2 \text{ s}, 0.8 \text{ s}] \quad (1)$$

Model Summary

Constant coefficients:				
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.7201	1.0344	-0.696	0.486
Smooth terms & functional coefficients:				
	edf	Ref.df	F	p-value
Intercept(yindex)	10.686	19.000	9.299	0.0237 *
ADHD(yindex)	2.001	2.378	0.251	0.8421
Female(yindex)	4.498	4.794	2.762	0.0159 *
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1				
$R^2_{\text{adj}} = 0.108$			Deviance explained = 11.1%	

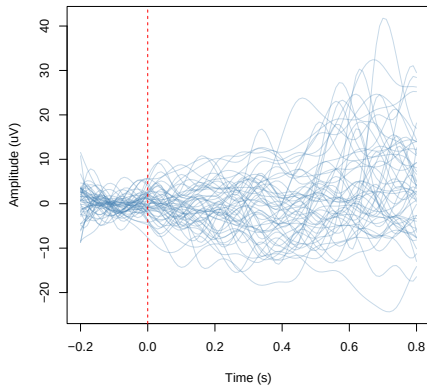
Table: Summary of the regression model.

Estimated Functional Regression Coefficients

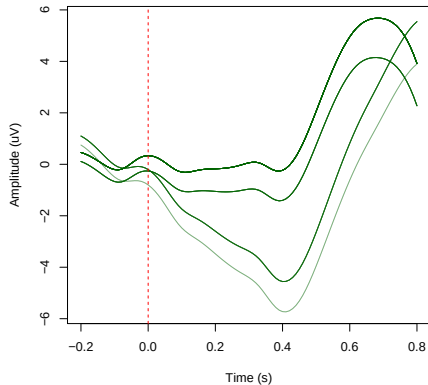


Observed and Fitted EEG Response Curves

Observed EEG Curves



Fitted EEG Curves



Conclusions

Conclusions

- **Between-subject variability** is concentrated in the post-stimulus window (0.4–0.8 s), with the first three functional principal components explaining 86.6% of total variance.
- **Function-on-scalar regression** extended the hypothesis testing by simultaneously modeling ADHD and gender effects on the EEG curves at channel FC1.
- **Gender** showed a statistically significant effect ($p = 0.016$), with female subjects exhibiting lower amplitudes in the post-stimulus window (0.3–0.6 s), while **ADHD** status had no significant effect ($p = 0.842$).
- The model's low explanatory power ($R^2 \approx 0.11$) indicates that substantial individual variability remains unexplained, suggesting other covariates may contribute to EEG response patterns.

Thank you for your time!